Quantum Computing for Traffic Optimization in Smart Cities

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1 Introduction

Urban traffic congestion remains a persistent challenge in modern cities, leading to increased travel times, fuel consumption, and environmental impact. Efficient traffic signal coordination and routing strategies are critical for mitigating these issues, yet they often require solving complex combinatorial optimization problems. Classical methods, while effective in many cases, can struggle to scale as network size and complexity increase. [2]

This project explores the use of hybrid quantum-classical algorithms for traffic optimization, with a focus on the Quantum Approximate Optimization Algorithm (QAOA). By modeling traffic signal coordination as a combinatorial optimization problem and encoding it into a Quadratic Unconstrained Binary Optimization (QUBO) formulation, we investigate whether quantum-inspired methods can provide competitive or improved solutions compared to classical baselines.

2 Background and Motivation

A big motivation comes from the rapid global shift toward smart city infrastructure, where transportation systems are increasingly designed as integrated, data-driven networks. Experimental developments such as Toyota Woven City [5] demonstrate how future urban environments may be built from the ground up with interconnected mobility systems, autonomous vehicles, and real-time traffic coordination. At the same time, existing urban centers like Singapore and Tokyo are actively transitioning toward smart infrastructure by deploying intelligent traffic management systems, sensor networks, and adaptive signaling

technologies. These developments significantly increase both the scale and complexity of traffic optimization problems, creating a strong need for new computational paradigms. In this context, quantum optimization methods offer a compelling direction for addressing large-scale combinatorial challenges in next-generation cities.

Traffic optimization problems are inherently combinatorial, involving discrete decisions such as signal phases, routing paths, or scheduling policies. These problems are commonly addressed using classical techniques such as greedy heuristics, simulated annealing, or integer programming. [7]

Recent advances in quantum computing have introduced new approaches for tackling such problems. In particular, QAOA is a hybrid algorithm that leverages parameterized quantum circuits combined with classical optimization to approximate solutions to QUBO problems [4]. While current quantum hardware is limited, quantum simulators allow for experimentation on small to medium-sized instances.

The motivation for this project is twofold. First, classical methods face scalability challenges as traffic networks grow in size and complexity [8]. Second, hybrid quantum algorithms may provide improved exploration of solution spaces through quantum effects such as superposition and interference.

3 Proposed Method / Design Details

3.1 Traffic Network Modeling

The traffic system is modeled as a graph where intersections are represented as nodes and roads as edges. Each edge is assigned a weight corresponding to traffic intensity. Random traffic networks are generated using Erdős–Rényi graphs [9] to enable controlled experimentation across different sizes.

3.2 Problem Formulation

The optimization task focuses on assigning traffic signal phases to each intersection. The objectives are:

- Enforce that each intersection selects exactly one phase
- Encourage coordination between neighboring intersections
- Improve overall traffic flow across the network

This problem is formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem [3]. The QUBO includes:

- One-phase constraints to ensure valid signal assignments
- Flow reward terms to promote efficient traffic movement
- Coordination terms to align neighboring intersections
- Balance penalties to avoid degenerate solutions

3.3 Quantum Optimization (QAOA)

The QUBO formulation is converted into a form compatible with Qiskit and solved using QAOA. This involves:

- Constructing parameterized quantum circuits
- Using classical optimizers such as COBYLA or SPSA [1]
- Sampling measurement outcomes to obtain candidate solutions

The implementation uses a circuit depth of $p = 1$ and runs on a quantum simulator.

3.4 Classical Baselines

Three classical approaches are implemented for comparison:

- Greedy algorithm
- Simulated annealing [6]
- Exact solver (for small instances)

3.5 Experimental Framework

A unified experimental pipeline is used:

1. Generate traffic graph
2. Construct QUBO
3. Solve using all methods

4. Record objective values and runtime

4 Evaluation Methodology

The methods are evaluated using the following metrics:

4.1 Solution Quality

Measured using the QUBO objective value. Lower values correspond to better traffic configurations. So a score of -176 would be more ideal than -90.

4.2 Runtime Performance

Execution time is recorded for each method to evaluate computational efficiency. Whilst this is useful to compare the different classical baselines we have, we must also rememebr that this is being run on a classical computer simulating a quantum computer, so the actual run time performances would not accurately reflect onto these results.

4.3 Scalability

Experiments are conducted for graph sizes of $n = 4, 6, 8, 10$ nodes. Exact solutions are computed only for small instances. ($n \leq 6$).

4.4 Convergence Behavior

For QAOA, convergence properties and sensitivity to optimization parameters are analyzed.

5 Experiments and Results

5.1 Small-Scale Validation

For small graphs ($n \leq 6$), QAOA produces near-optimal solutions when compared to the exact solver. Simulated annealing often matches or slightly outperforms QAOA, while greedy methods perform worse due to their local nature.

5.2 Medium-Scale Performance

For larger graphs ($n = 10$), simulated annealing achieves the best solution quality overall when considering time overhead. QAOA remains competitive but produces slightly

higher-cost solutions. However, QAOA exhibits more consistent performance across runs, both time wise, and solution wise, and QAOA had nearly exact or better performances compared to simulated annealing as n increased.

5.3 Runtime Analysis

Greedy methods are the fastest, followed by simulated annealing. QAOA has the highest runtime due to quantum circuit simulation overhead.

5.4 Impact of Multi-Phase Modeling

The multi-phase QUBO formulation improves realism by modeling real traffic signal systems more accurately. However, it also increases problem size and computational complexity.

5.5 Scalability Trends

As the network size increases:

- QAOA performance degrades gradually but remains stable
- Classical methods scale more efficiently
- QAOA maintains competitive solution quality

6 Conclusion

This project demonstrates that traffic signal optimization can be effectively modeled as a QUBO problem and solved using hybrid quantum-classical methods. QAOA achieves near-optimal performance on small instances and remains competitive on larger ones, although classical methods currently compare well = in efficiency and scalability when testing QAOA through classical simulation.

7 Future Work

Some future improvements and comparisons that I did not have the resources or time to to do could include:

- Increasing/Changing QAOA circuit depth

- Exploring noise-aware optimization techniques
- Testing on real-world traffic datasets (using existing sets or creating my own from scratch)
- Investigating hybrid decomposition methods
- Running experiments on real quantum hardware (Linked to the first option as well)

References

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